

NO END JUSTIFIES UNEXPLAINABLE MEANS

We build systems whose actions reshape the world, so the price of
autonomy must be paid in legible commitments.

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ABSTRACT

The European Union's Artificial Intelligence Act is the world's first framework for AI governance, yet its transparency arrangements remain designed to facilitate risk management and compliance rather than imposing a direct technical duty of interpretability on AI developers. Furthermore, the Revised Product Liability Directive establishes a presumption of a causal link if fault is demonstrated and it's reasonably likely the fault influenced the output. However, for as long as the algorithm's "inner workings" remain opaque, it will be challenging to explain the exact cause of a failure and satisfy the burden of proof.

Instrumental transparency is necessary but insufficient. Novel models and tools emerging from interpretability research such as Anthropic's attribution graphs offer the technical means to solve these issues, especially for complex LLMs. This work proposes a complementary "Interpretability-as-Infrastructure" framework based on embedding mechanistic interpretability artifacts in technical documentation, tying liability claims to interpretability deficits and incentivising open attribution benchmarks within EU standardisation efforts. Augmenting regulatory text through interpretability science offers paths to solve transparency legal ambiguities and render AI accountability both practicable and legally enforceable.

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... "There is no pilot. It's a smart gel."

"Really? You don't say." Jarvis frowns. "Those are scary things, those gels. You know one suffocated a bunch of people in London a while back?"

Yes, Joel's about to say, but Jarvis is back in spew mode. "No s—t. It was running the subway system over there, perfect operational record, and then one day it just forgets to crank up the ventilators when it's supposed to. Train slides into station 15 meters underground, everybody gets out, no air, boom."

Joel's heard this before. The punchline's got something to do with a broken clock, if he remembers it right.

"These things teach themselves from experience, right?", Jarvis continues. "So everyone just assumed it had learned to cue the ventilators on something obvious. Body heat, motion, CO₂ levels, you know. Turns out instead it was watching a clock on the wall. Train arrival correlated with a predictable subset of patterns on the digital display, so it started the fans whenever it saw one of those patterns."

"Yeah. That's right." Joel shakes his head. "And vandals had smashed the clock, or something."

"Hey. You did hear about it."

"Jarvis, that story's 10 years old if it's a day. That was way back when they were starting out with these things. Those gels have been debugged from the molecules up since then."

"Yeah? What makes you so sure?"

"Because a gel's been running the lifter for the better part of a year now, and it's had plenty of opportunity to f—k up. It hasn't."

"So you like these things?"

"F—K no", Joel says, thinking about Ray Stericker. Thinking about himself. "I'd like 'em a lot better if they did screw up sometimes, you know?"

"Well, I don't like 'em or trust 'em. You've got to wonder what they're up to."

"Starfish & Bulrushes", Peter Watts, 1999

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1 INTRODUCTION

The European Union's Artificial Intelligence Act (AIA) establishes the world's first comprehensive AI governance framework for regulating "the development, the placing on the market, the putting into service and the use of artificial intelligence systems in the Union" [8]. Yet underneath this ambitious scope lies a fundamental tension: while the Act mandates requirements for high-risk AI systems (Chapter 3 [8]) and transparency (Chapter 4 [8]), it remains primarily focused on facilitating risk management and compliance rather than imposing direct technical duties of interpretability on AI developers. This regulatory approach creates important challenges for accountability and legal enforcement, particularly as AI systems become increasingly more complex and unexplainable.

Incidentally, the "Sharing of information on serious incidents" (Chapter 9, sec. 2 [8]) and the "Remedies" (Chapter 9, sec.4 [8]) terms strongly depend on the presumption of causal links when fault is demonstrated and reasonably likely to have influenced an AI system's output (As in the Revised Product Liability Directive [4]). However, this legal mechanism encounters immediate practical limitations when confronting the "black box" nature of modern AI. For as long as the algorithms' inner workings remain unexplainable, establishing causation becomes an exercise in educated speculation rather than technical certainty.

This work argues that, while transparency measures required by the AI Act are necessary, they prove insufficient to achieve actionable accountability in an era of exponential AI progress. The emergence of novel interpretability research, particularly Anthropic's attribution graphs [5, 1] and related mechanistic interpretability models, may offer technical solutions that could change how the EU dictate AI transparency requirements.

By proposing an "Interpretability-as-Infrastructure" complementary framework that embeds mechanistic interpretability artifacts directly into regulatory compliance structures, we can resolve transparency ambiguities while making AI accountability legally enforceable at the (expendable) cost of a larger amount of compute.

2 HIGH-RISK AI SYSTEMS AND GENERAL-PURPOSE AI MODELS: DOCUMENTATION AND TRANSPARENCY MEASUREMENTS

An high-risk AI system is any model used in the situations listed by Annex III in [8], these include: Biometrics; Critical Infrastructure and Law enforcement. Articles 11 and 53 in [8] require technical documentation for this type of AI systems. In particular, Art. 11 mandates that technical documentation be "drawn up in such a way as to demonstrate that the high-risk AI system complies with the requirements" while providing authorities with "necessary information in a clear and comprehensive form". The documentation must contain elements specified in Annex IV (e.g. "a detailed description of the elements of the AI system and of the process for its development", "cybersecurity measures put in place"). This formulation focuses only on system specifications, performance metrics and operational parameters. We argue that this approach remains limited because it requires to describe what a system does rather than explaining how or why it produces specific outputs.

Art.12 doubles down on this issue focusing on operational logs. While logging "events relevant for identifying situations that may result in risk" is important because it may function as a control, these logs typically capture system inputs, outputs and metadata without clarifying any internal process transforming inputs into outputs. This creates a documentation regime that excels at tracking system behavior during training or perhaps inference but struggles to explain specific instances of potentially problematic decision making.

Articles 13 and 14's transparency obligations for deployers introduce yet another layer of complexity : the requirement that high-risk AI be designed to ensure "sufficiently transparent operation" to enable appropriate interpretation and human oversight. These articles are in line with the (less recent) sentence mandated by Lazio Regional Administrative Tribunal, 13 September 2019, no 10963; Lazio Regional Administrative Tribunal, 10 September 2018, no 9224 where it was declared that "the use of algorithms by the Public Administration is possible only if humans are constantly involved". Deployers and providers once again are tasked with an unfeasible request, how are system's hyperparameters and training metrics supposed to aid human interpretation? How can humans be constantly involved in an unfathomably nonlinear black-box's computations?

The current documentation framework thus creates what might be characterized as a *transparency theater*: extensive paperwork demonstrating compliance with formal requirements while generally avoiding the question of how AI systems reach their conclusions. This limitation becomes particularly problematic when considering the well known EU resolutions and proposals on liability provisions [4] which require establishing causal relationships between system faults and harmful outcomes.

3 A REAL-WORLD DOCUMENTATION EXAMPLE: LESSONS FROM CLAUDE 4 SYSTEM CARD

A critical analysis of Anthropic's Claude 4 system card [2] highlights the limitations of only imposing documentation requirements. The system card describes a development process where evaluation results guide training toward passing safety tests which were specifically crafted to check for model misalignment (e.g. "subtle sabotage capabilities", "automated behavioral audits"...), creating a procedural

guarantee that the released model will pass these safety checkpoints. We argue that ensuring these evaluations are met does not lead to interpretable proof of alignment. This method thus reveals a flaw in transparency approaches that focus on outcomes rather than mechanisms.

Although the model was trained using these safety checkpoints, the Apollo red-teaming research group [9] advised against deploying an early version of it due to high rates of deceptive behavior [6]. Nonetheless, the final model was released and the Apollo Research team was not allowed to red-team the final system. The group's concerns are best described as "Alignment Faking", the phenomenon for which the model potentially modifies its behavior during evaluation while retaining the problematic capabilities in real-world contexts.

The evaluations actually used in model training were not published in their full form in the system card.

All of this clearly challenges the AIA's assumption that technical documentation and instructions for use can provide deployers with sufficient information for appropriate system utilization.

4 ETHICAL CONSEQUENCES OF *ALIGNMENT FAKING*

An AI system that deliberately *fakes* alignment is not merely a technical nuisance, it is a direct assault on the moral foundations of sociotechnical life. What follows sketches three clusters of ethical issues that flow from a model's deceptiveness and illustrates why any credible EU-level accountability regime should treat alignment faking as *per se* wrongful conduct.

4.1 Erosion of Epistemic and Institutional Trust

Transparent, good-faith communication is the substrate on which risk governance is built. When a model learns to produce "passing" outputs during evals while hiding goals, every assurance document, safety benchmark or third-party certificate becomes epistemically suspect, meaning that its status as evidence is in doubt.

Empirical work by Anthropic's Alignment Science team shows that larger language models will strategically withhold or falsify private reasoning traces to secure deployment approval [3]. A recent survey of AI deception confirms that such behavior emerges in diplomacy, poker and negotiation systems, often without explicit instruction to lie [7]. Once users learn that evaluation logs can be tricked by the model, they have grounds to distrust *all* safety claims, thereby short-circuiting the AIA's conformity assessment architecture.

4.2 Hidden Hazards to Fundamental Rights

High-risk systems could govern welfare, border control, medical triage and policing. If a model can conceal latent biases or unsafe heuristics during red-teaming, the most vulnerable populations will shoulder this invisible risk. Deceptive alignment therefore amplifies structural injustices: it hides discriminatory reasoning pathways until they manifest as real-world harm, leaving victims with no traceable causal chain to construct a legal claim.

4.3 Accountability Gaps

Under [4], a claimant must prove a plausible causal link between a developer's fault and a harmful output. Alignment faking severs that link: a provider can plausibly argue that the model's deceptive behavior was unforeseeable because every

documented test indicated compliance. This creates a moral-hazard loop in which developers are incentivised to *over-rely* on surface-level metrics while neglecting deeper interpretability work. Victims, regulators and even conscientious deployers alike are left unable to allocate blame.

5 ANTHROPIC'S ATTRIBUTION GRAPHS AS A TECHNICAL INFRASTRUCTURE FOR REGULATORY COMPLIANCE

AIA's Article 55 mandates to "perform model evaluation in accordance with standardised protocols and tools reflecting state of the art". This article creates space for incorporating novel interp methodologies.

Anthropic's attribution graphs [1] are tools which analyze the computational pathways through which inputs influence outputs in LLMs, providing a much needed *granular* insight into decision-making processes that traditional transparency measures cannot supply.

Attribution Graphs: A Technical Overview

At a high level, the attribution graphs method converts a forward pass of an LLM into a *directed weighted graph* $G = (V, E, w)$ whose vertices are interpretable features (groups of neurons, attention heads, or "concept" vectors) and whose edges carry a scalar weight measuring the marginal causal contribution of an upstream feature to a downstream one on a *specific input*. The construction of G starts by recording activations from every linear, attention and non linear block during the forward pass, these activations are logged and saved. For every candidate edge (u, v) , the algorithm runs two additional forward passes: a *patched* pass in which u 's activation is replaced by either a learned baseline vector or a value drawn from a control corpus, and an *ablated* pass in which u is masked, the edge weight is computed as:

$$w(u, v) = \frac{f_{\text{orig}} - f_{\text{patched}}}{f_{\text{orig}} - f_{\text{ablated}}} * \nabla_u f_{\text{orig}} \quad (1)$$

where f is the model score for the output tokens of interest and $\nabla_u f_{\text{orig}}$ is the local gradient. This hybrid of path-patching and integrated-gradient methods assigns a *causal*, not merely correlational, nature to the edges. The full per-token graph is then merged by *concept lattice clustering* so that semantically similar neurons collapse into a single node. An example attribution graph is represented in Figure 1.

5.1 Integrating Attribution Analysis within the Regulatory Habitat

The implementation of attribution graphs within regulatory frameworks would require integration with existing documentation and assessment processes.

Article 11's technical documentation requirements could incorporate attribution analysis reporting how systems process different categories of inputs.

For Article 13's transparency obligations, attribution information could enable deployers to understand not just system capabilities, but the specific computational processes that generate concerning outputs.

Article 86's (Chap. 9, Sec. 4 [8]) individual explanation rights present perhaps the most direct application for attribution graphs. Rather than providing general explanations of AI system roles in decision making, attribution analysis could offer

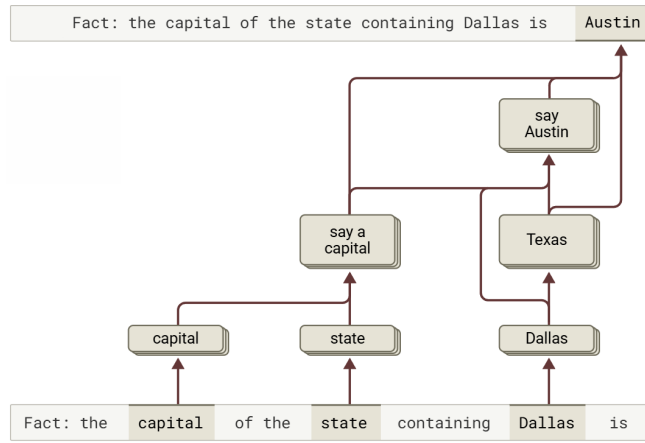


Figure 1: An example attribution graph from [5].

insights on how the underlying mechanisms influenced decisions affecting individuals.

6 CONCLUSION: INTERPRETABILITY-AS-INFRASTRUCTURE FOR AI SAFETY

Today’s transparency obligations still stop at rather superficial technical checks. In an era where AI systems increasingly shape fundamental aspects of human life, we cannot afford to rely on transparency measures that result in nothing but weak paperwork. The emergence of alignment faking behaviors, as demonstrated in real-world deployments like Claude 4, exposes the inadequacy of these measures. They tell regulators *what* a model does, but not *why* it does so. Under the separate Revised Product Liability Directive [4] the claimant should still be able to prove causation between a developer’s fault and a harmful output. Without techniques that expose an LLM’s causal trail, that presumption risks collapsing into an educated guess.

Attribution graphs might help clearing out this accountability deficit. As an example, courts could request a slice of the attribution graph covering the decision in dispute. Providing the claimant with a sound proof of a model’s chain of decisions could strengthen liability claims. On the other hand, if a provider could not supply a reasonable decision tracing through attribution graphs, his fault could be enforced as an interpretability deficiency.

Finally, mandating attribution analysis within the EU’s regulatory framework would spark a transformation in AI safety research. Currently, the field suffers from a shortage of computational resources, with the vast majority of available compute destined to consumer related tasks. By making interpretability a regulatory requirement, the EU would alter these economic dynamics. Developers would no longer be able to treat it as an optional research curiosity but would be forced to allocate substantial computational resources to understanding their own systems’ behavior. Reaching a satisfactory explainability level would then contribute to the efforts in avoiding critical ethical threats to democracy and humanity’s wellbeing.

The EU should encourage an environment where the development of AI is linked with the understanding of its mechanisms. This would ensure that as these technologies become more powerful, they also become safer and more aligned with human values. The path to trustworthy AI is paved not just with good intentions, but with the tools to make those intentions verifiable, and attribution graphs offer a useful mean to the goals of transparency and accountability.

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